

Rakshya U. Sharma

Curriculum Vitae

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Research Profile

Computational geneticist and bioinformatician with nearly five years of progressive research experience (2021–present) in statistical genomics and large-scale human genetic analysis. Develops scalable, reproducible pipelines and analytic frameworks for polygenic risk modeling and Mendelian randomization across multi-cohort datasets. Expertise in workflow engineering (Snakemake/Nextflow), high-performance computing, and rigorous statistical modeling. Demonstrated leadership in computational infrastructure development, trainee mentorship, and contributions to peer-reviewed research in complex disease genetics.

Publications and Pre-prints

5. Meri Okorie, Xiaqing Jiang, Paulina Tolosa-Tort, **Rakshya U. Sharma**, Alexandra L. Clark, Kristine Yaffe, View ORCID ProfileJennifer S. Yokoyama, View ORCID ProfileShea Andrews, Health and Aging Brain Study - Health Disparities. **"Orthogonal Contributions of Genetic, Clinical, and Social Determinants of Health Risk Burdens on Alzheimer's Disease Pathophysiology"**. July 2026. DOI: <https://doi.org/10.64898/2026.07.07.26357509>.
2. Shea J Andrews, **Rakshya U Sharma**, Brendan A Mitchell, Tong Tong, Luke W. Bonham, Alan E Renton, View ORCID ProfileXiaoling Zhang, Marina Sirota, Duygu Tosun, Kristine Yaffe, the Alzheimer's Disease Neuroimaging Initiative. **"Integrative effects of Telomere Length, Epigenetic Age, and Mitochondrial DNA abundance in Alzheimer's Disease"** Pre-print *medRxiv*, September 2025. DOI: <https://doi.org/10.1101/2025.07.16.25331683>
3. Paulina Tolosa-Tort, Meri Okorie, Aadrita Chatterjee, Ana I Borieu, **Rakshya U. Sharma**, Shea J Andrews. **"Toward Personalized Medicine in AD/ADRD Through Genetic-Exposome Dementia Risk Assessments."** *npj Dementia*, September 2025
4. Aadrita Chatterjee, Brian Alvarez Alvarez, **Rakshya U Sharma**, Caroline Jonson, Heather M Wilkins, Judy Pa, Russell H Swerdlow, Kristine Yaffe, Alison M Goate, Shea J Andrews, Alzheimer's Disease Genetics Consortium, Alzheimer's Disease Neuroimaging Initiative. **"Evaluating the causal effect of mitochondrial dysfunction on Alzheimer's and Parkinson's disease using Polygenic Risk Scores and Mendelian Randomization"** Pre-print *medRxiv*, September 2025. DOI: <https://doi.org/10.1101/2025.09.25.25336251>

5. Ben Weisburd, **Rakshya Sharma**, Villem Pata, Tiia Reimand, Vijay S. Ganesh, Christina Austin-Tse, Ikeoluwa Osei-Owusu, Emily O'Heir, Melanie O'Leary, Lynn Pais, Seth A. Stafki, Audrey L. Daugherty, Chiara Folland, Stojan Peric, Nagia Fahmy, Bjarne Udd, Magda Horáková, Anna Łusakowska, Rajanna Manoj, Atchayaram Nalini, Veronika Karcagi, Kiran Polavarapu, Hanns Lochmüller, Rita Horvath, Carsten G. Bönnemann, Sandra Donkervoort, Göknur Haliloğlu, Ozlem Herguner, Peter B. Kang, Gianina Ravenscroft, Nigel Laing, Hamish S. Scott, Ana Töpf, Volker Straub, Sander Pajusalu, Katrin Öunap, Grace Tiao, Heidi L. Rehm, Anne O'Donnell-Luria. **"Diagnosing missed cases of spinal muscular atrophy in genome, exome, and panel sequencing data sets"** *Genetics in Medicine*. April 2025. DOI: <https://doi.org/10.1016/j.gim.2024.101336>.

Manuscripts Under Preparation

2. **Rakshya U Sharma**, Meri Okorie, Aadrita Chatterjee, Caroline Jonson, Kristine Yaffe, Michaël E Belloy, Shea J Andrews for ADGC and HABS-HD Study team **"Sex-Specific Genetic and Causal Effects of Modifiable Risk Factors on Alzheimer's Disease: Evidence from PRS and Mendelian Randomization."** *npj Dementia*, TBD

Conference Poster Presentations

1. **Rakshya U. Sharma**, Meri Okorie, Nia Spadoni, Aadrita Chatterjee, Kristine Yaffe, Shea J. Andrews for the Alzheimer's Disease Genetics Consortium. **"Polygenic Risk Scores for modifiable risk factors show sex-specific associations in Alzheimer's Disease"** *2025 Alzheimer's Association International Conference, Toronto, Canada*
2. Aadrita Chatterjee, Brian Alvarez, **Rakshya U Sharma**, Heather Wilkins, Russell H Swerdlow, Kristine Yaffe, Shea J Andrews. **Evaluating the Causal Effect of Mitochondrial Dysfunction on Alzheimer's and Parkinson's Disease using Polygenic Risk Scores and Mendelian Randomization"** *2025 Alzheimer's Association International Conference, Toronto, Canada*
3. Meri Okorie, Xiaqing Jiang, PhD, **Rakshya U. Sharma**, Caroline Jonson, PhD, Kristine Yaffe, PhD, Jennifer S. Yokoyama, PhD, Shea J. Andrews, PhD, for Healthy Aging Brain Study – Health Disparities. **"Orthogonal Contributions of Genetic and Clinical Risk Factors on Alzheimer's Disease and Neurodegenerative Pathogenesis"** *2025 Alzheimer's Association International Conference, Toronto, Canada*
4. **Rakshya U. Sharma**, Dennis Mulligan, Alexis Thornton, Angela N. Brooks **"Building RNA Binding Protein Regulatory Signatures to Identify RNA Splicing Alterations in Cancer"** *2024 National Human Genome Research Institute (NHGRI) Annual Meeting, Seattle, WA*
5. **Rakshya U. Sharma**, Dennis Mulligan, Alexis Thornton, Angela N. Brooks **"Building RNA Binding Protein Regulatory Signatures to Identify RNA Splicing Alterations in Cancer"** *2023 Advances in Genome Biology and Technology-Precision Health meeting, Annual Biomedical Research Conference for Minoritized Scientists, San Diego, CA*
6. **Rakshya U. Sharma**, Dennis Mulligan, Alexis Thornton, Angela N. Brooks **"Building RNA Binding Protein Regulatory Signatures to Identify RNA Splicing Alterations in Cancer"** *2023 Annual Biomedical Research Conference for Minoritized Scientists (ABRCMS), Phoenix, AZ*

7. **Rakshya U. Sharma**, Dennis Mulligan, Alexis Thornton, Angela N. Brooks **"Identifying the RNA Splicing Patterns in Cancer using Mutually Exclusive Splicing Analysis"** *2023 National Human Genome Research Institute Annual Meeting, Salt Lake City, UT*
8. **Rakshya U. Sharma**, Dennis Mulligan, Alexis Thornton, Angela N. Brooks **"Identifying the RNA Splicing Patterns in Cancer using Mutually Exclusive Splicing Analysis"** *2022 SACNAS NDiSTEM Conference, San Juan, Puerto Rico*
9. **Rakshya U. Sharma**, Ben Weisburd, Heidi L. Rehm **"Spinal Muscular Atrophy Diagnosis Using Whole Exome Sequencing Data"** *2022 Broad Summer Research Program Poster Symposium, Cambridge, MA*
10. **Rakshya U. Sharma**, Laura R. Nichols, Daniela Ushizima **"Image Analysis of Biofuel Plant Root Composition in Drought Conditions"** *2021 Lawrence Berkeley National Lab Poster Symposium, Virtual*

Research Experiences

06/2024–
Present **Bioinformatician**, Andrews Lab, University of California, San Francisco, Department of Psychiatry and Behavioral Sciences, San Francisco, CA, PI: Shea Andrews Ph.D.

Funded by NIH-NIA and Alzheimer's Association

Project: Investigating Sex-stratified Causal Relationships Between Modifiable Risk Factors and Alzheimer's Disease utilizing Alzheimer's Diseases Genetics Consortium and Health Aging (ADGC) and Brain Study - Health Disparities (HABS-HD) datasets.

Statistical and Genomic Analysis

- Performed large-scale statistical modeling in R (logistic regression, stratified analysis) to evaluate how polygenic risk scores (PRS) for education, cholesterol, and other factors contribute differently to Alzheimer's disease risk in men and women.
- Produced visual summaries (forest plots, regression effect sizes, stratified confidence intervals) using ggplot2, forestplot, and RMarkdown, enabling clear interpretation and communication of results.
- Built expertise with statistical genetics software (PLINK, PRS-CS, pgsc_calc), expanding analytic toolkit to support multi-cohort analyses.

Pipeline Engineering and HPC Integration([GitHub: AndrewsLabUCSF/snakemake-executor-plugin-sge-wynton](#))

- Designed and implemented a custom Snakemake executor plugin to streamline job submission on UCSF's SGE-based Wynton cluster, resolving bottlenecks in storage and queueing.
- Optimized complex workflows using Snakemake and Nextflow, improving reproducibility, scalability, and runtime efficiency across multiple bioinformatics projects.
- Containerized pipelines with Docker, ensuring reproducibility and portability across diverse computing environments (HPC and cloud).

Collaboration, Mentorship, and Communication

- Mentored a ChangeSF summer intern (onboarding, code review, documentation, weekly check-ins).
- Presented updates in lab meetings and contributed to manuscripts/figures.
- Collaborated across disciplines to troubleshoot pipelines and integrate best practices.

04/2022– **Undergraduate Research Assistant**, *Brooks Lab, UCSC Genomics Institute*, Santa Cruz, CA, PI: Angela Brooks Ph.D.

Funded by NHGRI Research Mentoring Internship

Project: Building RNA binding protein (RBP) regulatory signature dataset to identify the genetic basis of RNA splicing alterations in cancer.

- Designed and implemented *RBPSig* pipeline (STAR, MESA) to generate RBP knockdown signatures from ENCODE data and query TCGA lung adenocarcinoma samples.
- Automated large-scale RNA-seq data acquisition, alignment, and splicing quantification, building a resource for pan-cancer analysis of RBP dysregulation.
- Validated pipeline by recapitulating known SUGP1/SF3B1 splicing overlap in LUAD (Alsafadi et al. 2020), demonstrating rigor and feasibility of linking RBP loss to cancer phenotypes.
- Applied statistical modeling (beta distribution of splicing junctions, multiple testing corrections) and integrated clinical databases (cBioPortal) to connect molecular signatures with patient-level genomic data.
- Developed skills in statistical modeling, multivariate analysis, and high-performance computing, while strengthening scientific communication through proposal writing and literature review.
- Presented project at UCSC Research Mentoring Symposium and external conferences, strengthening ability to communicate interdisciplinary genomics research.

05/2022– **Broad Summer Research Internship**, *Translational Genomics Group, Broad Institute of MIT and Harvard*, Cambridge, MA, PI: Heidi Rhem Ph.D.

Funded by NHGRI Broad Summer Research Program

Project: Spinal Muscular Atrophy Diagnosis Using Whole Exome Sequencing Data.

- Developed a pipeline (**GitHub:** *broadinstitute/sma-finder*) to detect Survival Motor Neurons deletions and mutations in whole-exome sequencing data, contributing to the identification of new diagnostic candidates for Spinal Muscular Atrophy (SMA).
- Applied genomic visualization tools like Integrative Genomics Viewer (IGV) to analyze and present complex data sets, improving data interpretability and supporting robust scientific conclusions.
- Research led to a peer-reviewed publication (listed in Publications).

06/2021– **Community College Internship**, *IDEAL team, Lawrence Berkeley National Lab*, Berkeley, CA, PI: Daniela Ushizima Ph.D.

Funded by DOE Office of Science's Office of Workforce Development for Teachers and Scientists (WDTS):

Project: Image Analysis of Biofuel Plant Roots Composition in Drought Conditions.

- Applied machine learning techniques to automate segmentation in plant root composition studies, enhancing efficiency and accuracy in identifying regions of interest to form masks.
- Designed and showcased research progress through a poster at the Lawrence Berkeley National Lab Poster Symposium 2021, illustrating the application of machine learning in bioinformatics.

Research Talks

2025 **2025 MAC at Large meeting, UCSF**, "*Sex differences in how genetic liability to modifiable traits influence Alzheimer's Diseases Risk*", Lightning talk.

2025 **PROPEL Scholars Symposium, UCSF**, "*Polygenic Risk Scores for modifiable risk factors show sex-specific associations in Alzheimer's Disease*".

Mentoring Experience

01/2025– **Research Mentor** to Mentee - Nia Spadoni, Community Health Advancing Next Generation Efforts San Francisco (CHANGE SF), University of California, San Francisco

08/2023– **Peer Mentor**, Services for Transfer and Re-entry Students (STARS), University of
06/2024 California, Santa Cruz

Memberships

06/2024– **ISTAART**, *International Society to Advance Alzheimer's Research and Treatment*.
Present

06/2022– **SACNAS**, *Society for Advancements of Chicanos/Hispanics and Native Americans in*
06/2024 *Science*.

Technical Skills

Programming **Python** (advanced), **R**, C++, C, Shell Script (Linux), LaTeX, Haskell, HTML, CSS
Languages

HPC SGE, SLURM, Amazon Web Services (AWS), Google Cloud Platform (GCP)

Workflow **Snakemake**, **Nextflow**, Git, Jupyter, Google Colab, various data analysis tools
management

Visualization Proficient with Integrative Genomics Viewer (IGV), UCSC Genome Browser, Applied
Tools Visualization and Analysis of Scientific Data using Python

Frameworks Docker, Anaconda, Machine Learning, Deep Learning, Artificial Intelligence

Education

06/2024 **Bachelor in Science in Computer Science, Minor in Bioinformatics**,
University of California, Santa Cruz, CA, **Graduation:** June 2024.

Trainings

07/2024 – **Post-Baccalaureate Research Opportunity to Promote Equity in Learning (PRO-PEL)**,
Present *University of California, San Francisco, CA*.

Participated in UCSF's PROPEL postbaccalaureate program, gaining formal training in biomedical data science, research communication, peer mentorship and critical literature analysis.

07/2021 – **Undergraduate Research Mentoring Internship (RMI), Genomics Institute**,
06/2024 *University of California, Santa Cruz, CA*.

Participated in UCSC Genomics Institute RMI program, gaining formal training in scientific research, professional development guidance and mentorship

Awards and Honors

2023–2024 Dean's Honors

2023 AGBT Precision health conference scholarship

2022 Koret Scholarship, UC Santa Cruz, Koret Foundation

2021–2024 NHGRI/NIH Undergraduate Research Mentoring Internship Scholarship, UC Santa Cruz,
Genomics Institute